AIAA 2205: Introduction to AI
Regression and Classification

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Contents to be fully mastered in this session

- ✓ Dataset (Training, Test, Validation)
- ✓ Regression (Concept and Application)
- ✓ Classification (Concept and Application)
- ✓ Overfitting and Underfitting
- ✓ Evaluation Metrics (ROC, AUC)

Content about this course

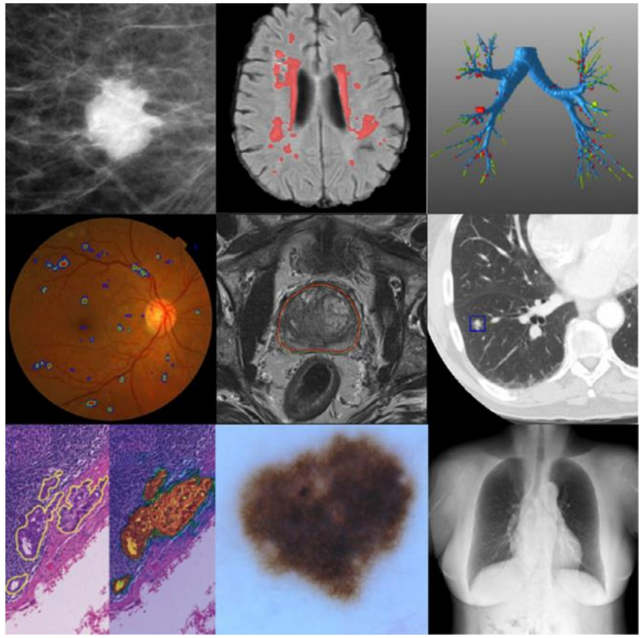
- ✓ **Dataset**
- ✓ **Regression**
- ✓ **Classification**
- ✓ **Overfitting and Underfitting**

Dataset

- What data is good for Deep Learning?
- Is dataset size larger the better?
- Some papers' main contribution is to build a new dataset, like the famous ImageNet by Feifei Li.

1000 object classes
(*i.e.*, cat, dog, flowers)

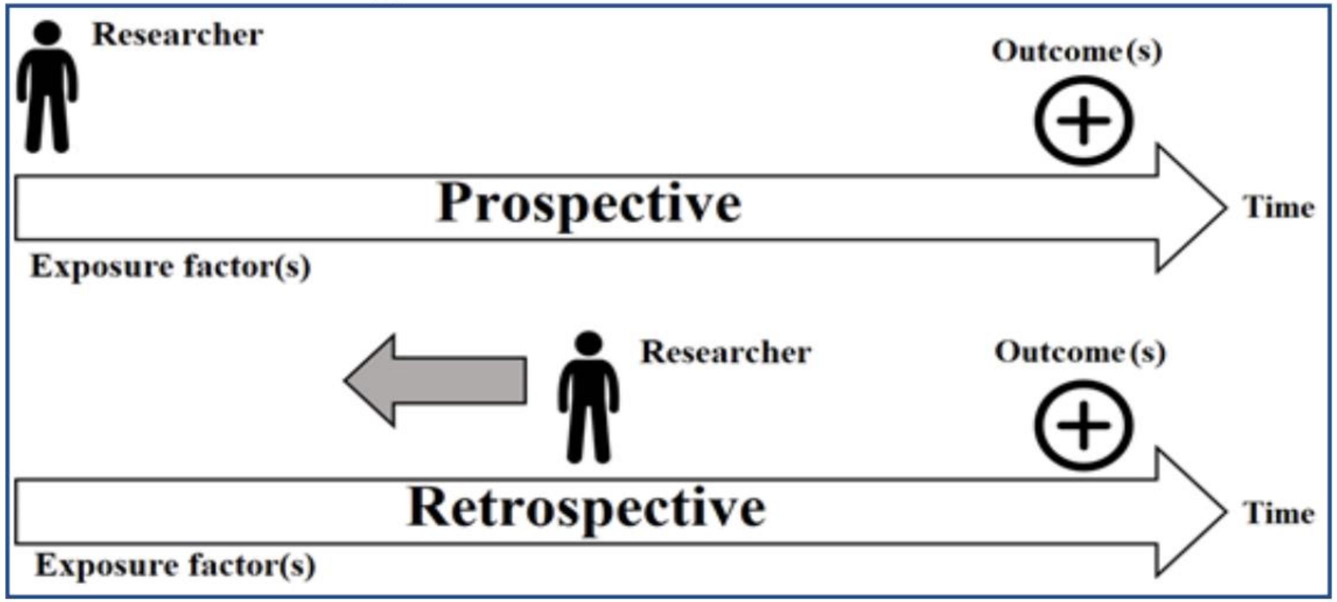
More than 14 million images in **ImageNet**



Around 100 - 10000 images per dataset

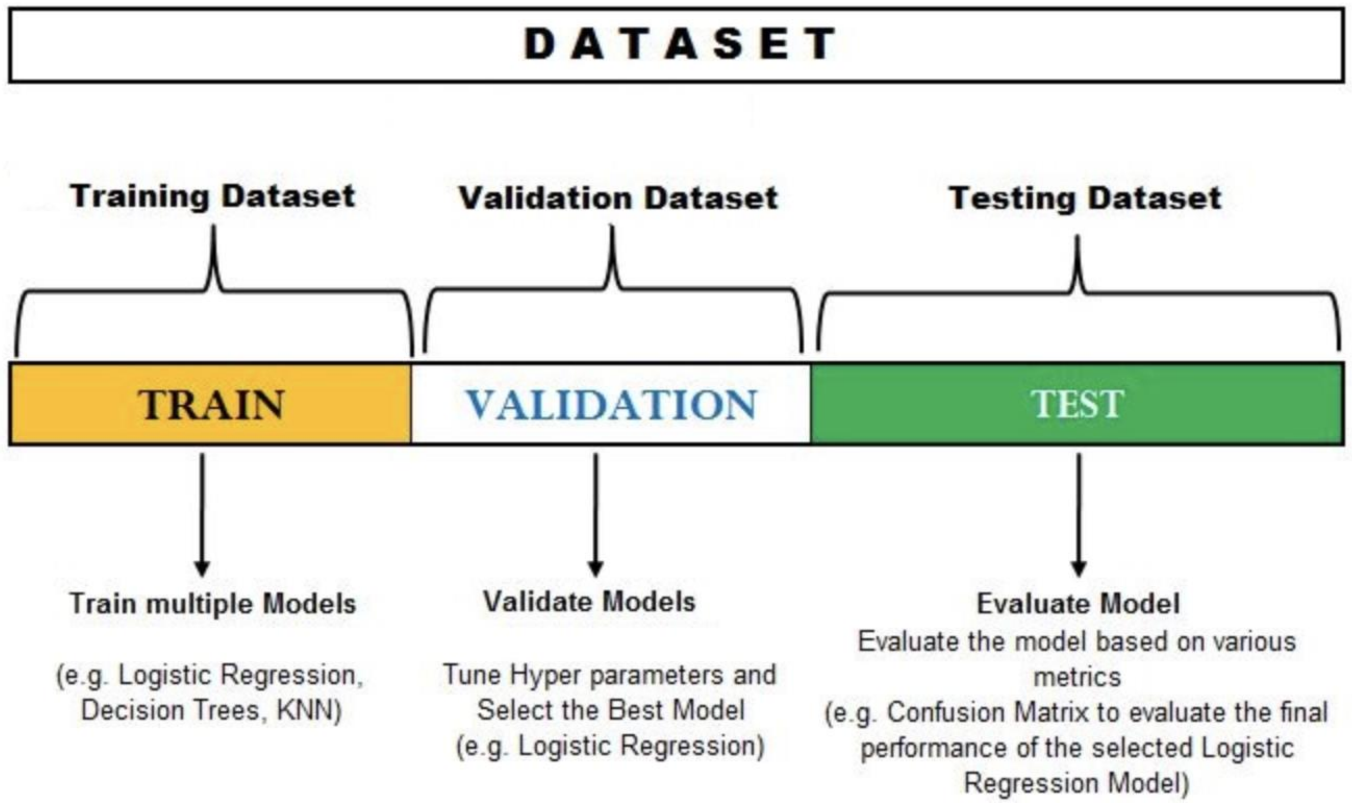
Retrospective *vs.* Prospective Data

- **Prospective Data** (前瞻性数据) refers to data that is collected in real-time. It is collected with the intention of analyzing it later to make predictions or understand patterns.
- **Retrospective Data** (回顾性数据) refers to data that is collected and analyzed after an event has occurred or a study has concluded. This data is often used for historical analysis and looking back to draw conclusions or insights.



Dataset

- Split to training set (or training set and validation set) and test set.



a. **Validation set**: the model occasionally "***sees***" this data, but never "***learn***" from this.
b. Based on validation set results, we update hyperparameters.
c. If the dataset is small, we can only split the dataset to training and test set.

k-fold Cross-Validation

- Estimating the ability of the model on new data.
- The procedure has a single parameter called k.
- k refers to the number of groups that a given data sample is to be split into.

k-fold Cross-Validation

- Shuffle the dataset randomly

1. Split the dataset into *k* groups
2. For each unique group:
    - Take the group as a hold out or test data set
    - Take the remaining groups as a training data set
    - Fit a model on the training set and evaluate it on the test set
    - Retain the evaluation score and discard the model
3. Summarize the skill of the model using the sample of model evaluation scores (mean and std)

An Example

Suppose we have a data sample with 6 observations:

```
1 [0.1, 0.2, 0.3, 0.4, 0.5, 0.6]
```

- The first step is to pick a value for k in order to determine the number of folds used to split the data.
- Here, we will use a value of $k = 3$. That means we will shuffle the data and then split the data into 3 groups.
- Because we have 6 observations, each group will have an equal number of 2 observations.

An Example

For example:

```
Fold1: [0.5, 0.2]
Fold2: [0.1, 0.3]
Fold3: [0.4, 0.6]
```

Three models are trained and evaluated with each fold given a chance to be the held out test set.

For example:

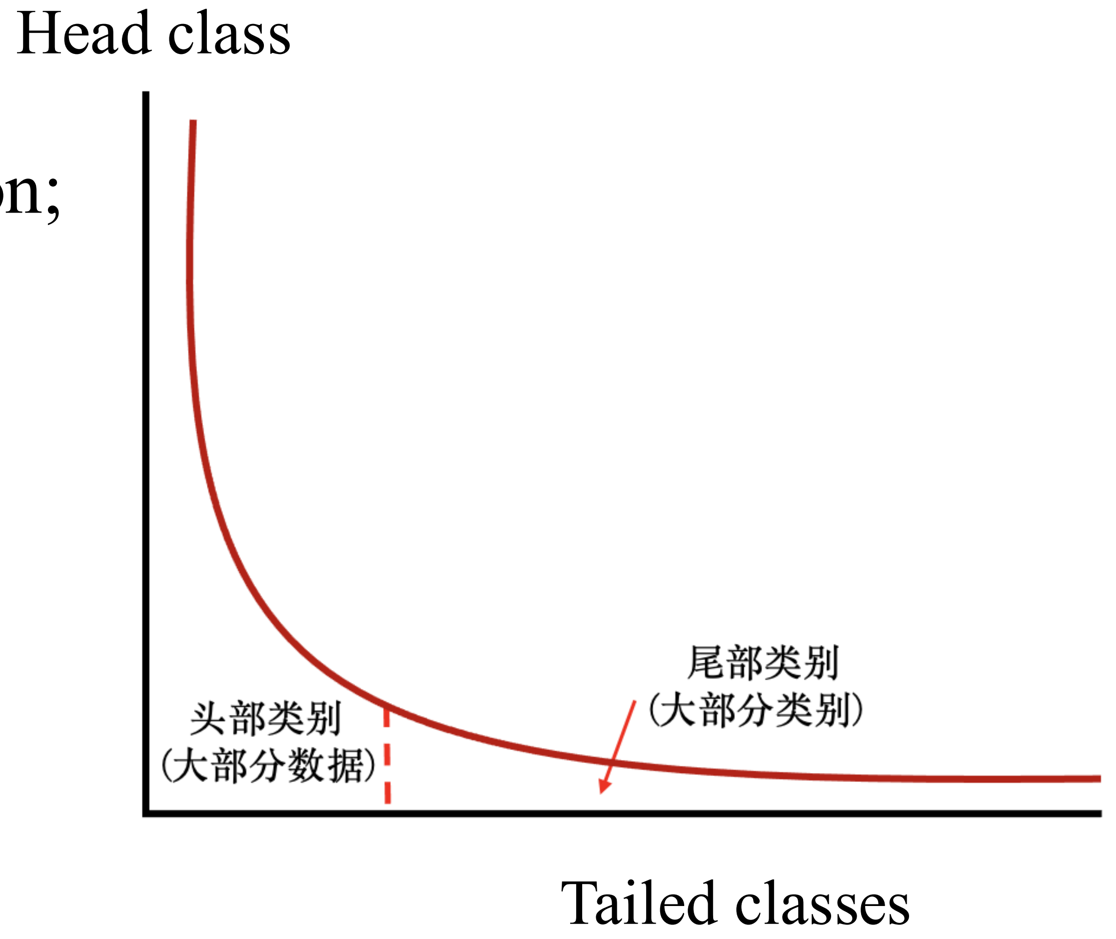
- **Model 1**: Trained on Fold1 + Fold2, Tested on Fold3
- **Model 2**: Trained on Fold2 + Fold3, Tested on Fold1
- **Model 3**: Trained on Fold1 + Fold3, Tested on Fold2

Taking the average of these three models as well as the *std* as the final result.

Dataset Challenges

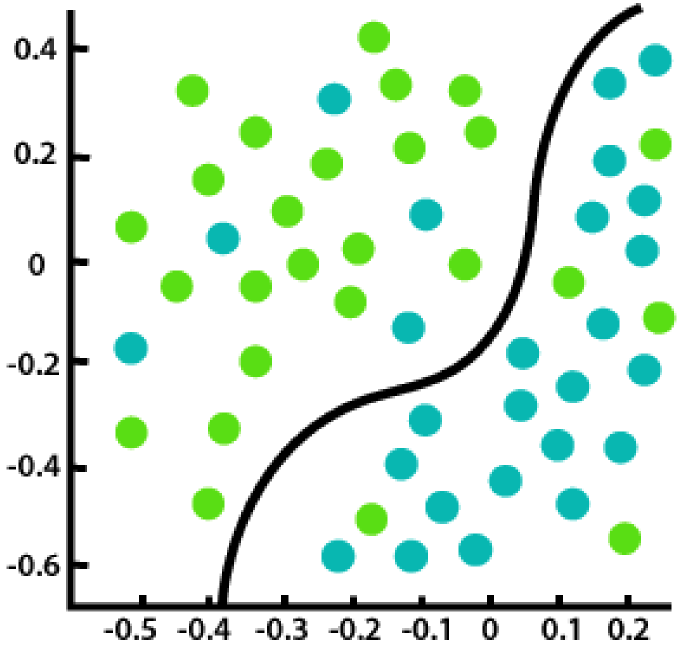
Recently, there are some **hot research topics** on data related with the performance of the Deep Learning based models:

1) Collect large-scale dataset;
2) Imbalanced (Long tailed) data distribution;
3) Weakly labelled data, Noisy label;
4) Useless data, noisy data, simple samples and hard samples.

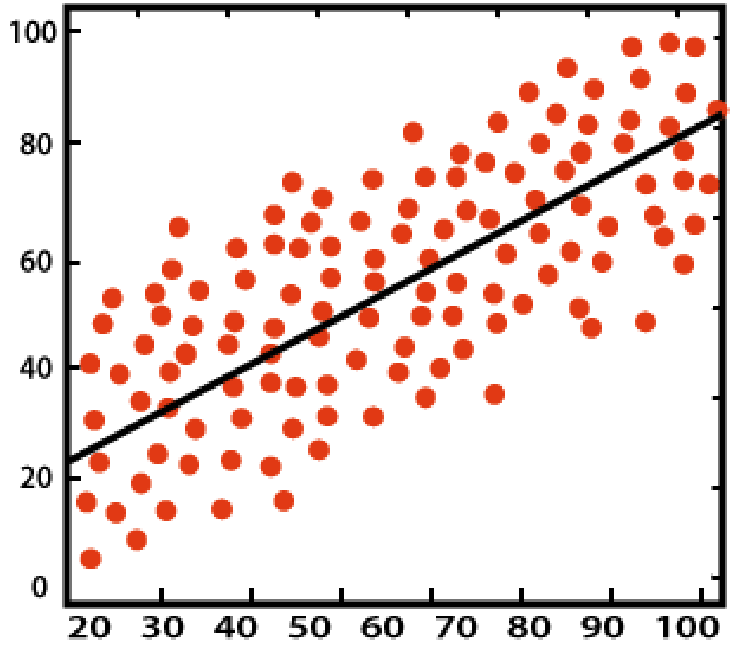


Difference between Classification and Regression

- **Classification**: used to predict/classify the **discrete** values such as Male or Female, True or False, Spam or not Spam, *etc*.
- **Regression**: used to predict the **continuous** values such as price, salary, *etc*.



Classification



Regression

Regression *vs.* Classification Examples

Suppose that you are running a company, and you want to develop learning algorithms to address the following two problems.

Problem 1: You have a large inventory of items. You want to predict how many of these items will sell over the next 3 months.

Problem 2: You'd like a program to examine individual customer accounts, and for each account, decide if it has been hacked or compromised.

Should you treat these as classification or as regression problems?

1. Treat both as classification problems.
2. Treat problem 1 as classification, problem 2 as regression.
3. Treat problem 1 as regression, problem 2 as classification.
4. Treat both as regression problems.

Linear Regression

Regression

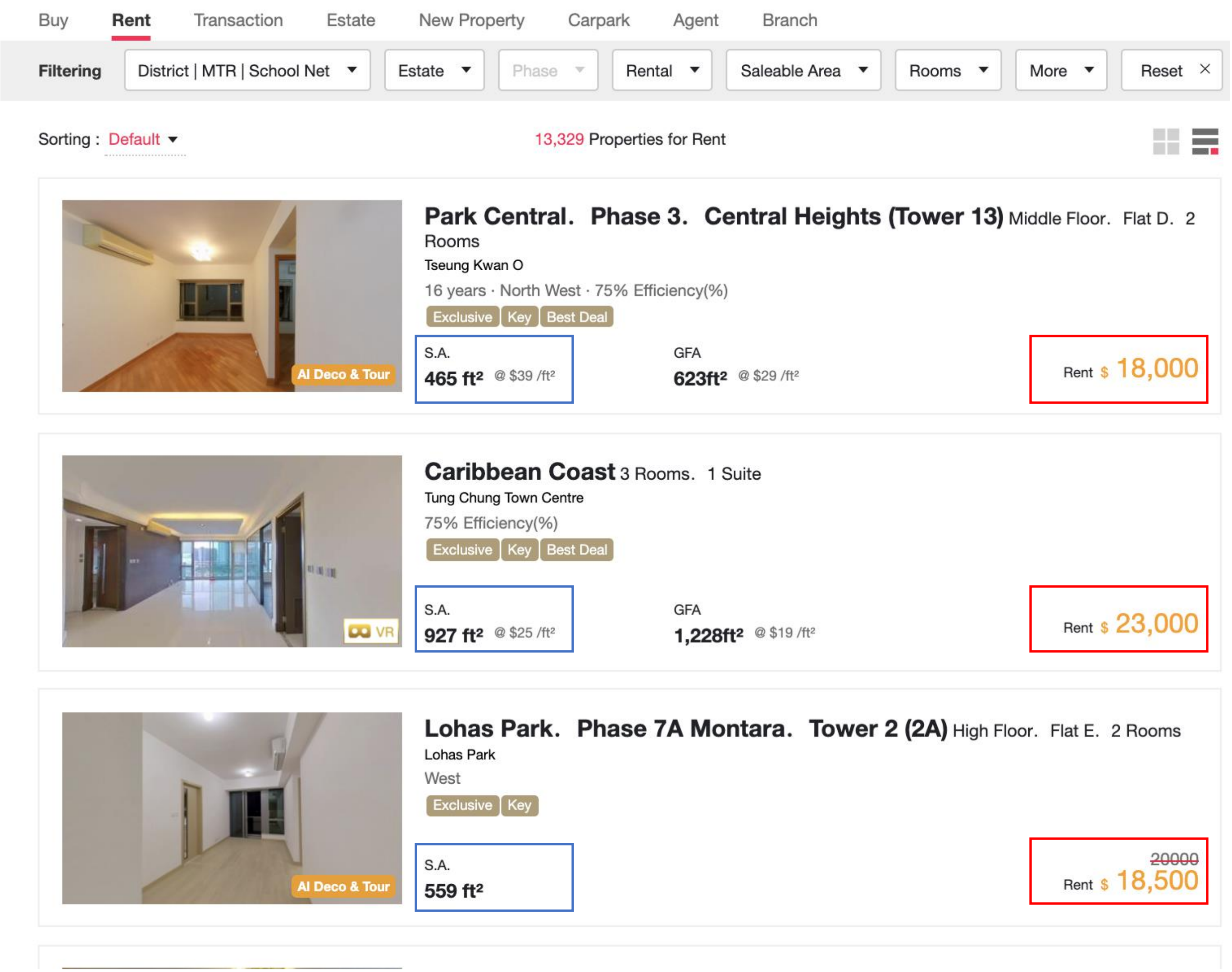
- The task of the regression algorithm is to find the mapping function to map the input variable (x) to the **continuous** output variable (y).
- **Example:** Suppose we want to do the weather forecasting (*e.g.*, temperature), so for this, we will use the regression algorithm.
- **Some Types of Regression Algorithms:**
  - Linear Regression
  - Support Vector Regression
  - Decision Tree Regression
  - Random Forest Regression

Common used loss functions:

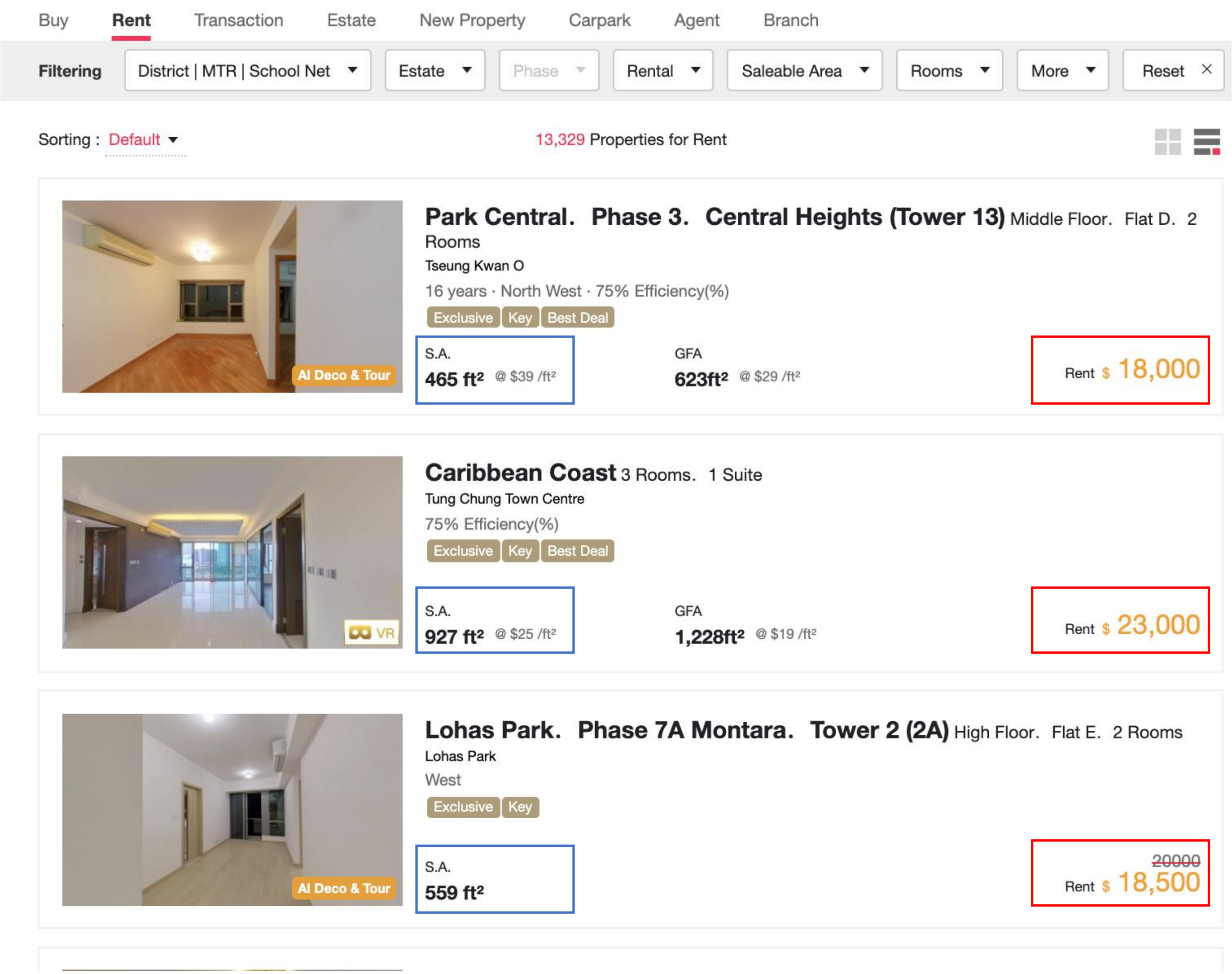
$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

Regression


Buy
Rent
Transaction
Estate
New Property
Carpark
Agent
Branch
Filtering
District | MTR | School Net
Estate
Phase
Rental
Saleable Area
Rooms
More
Reset
Sorting : Default
13,329 Properties for Rent
Park Central. Phase 3. Central Heights (Tower 13) Middle Floor. Flat D. 2 Rooms
Tseung Kwan O
16 years · North West · 75% Efficiency(%)
Exclusive
Key
Best Deal
AI Deco & Tour
S.A.
465 ft² @ $39 /ft²
GFA
623ft² @ $29 /ft²
Rent $ 18,000
Caribbean Coast 3 Rooms. 1 Suite
Tung Chung Town Centre
75% Efficiency(%)
Exclusive
Key
Best Deal
VR
S.A.
927 ft² @ $25 /ft²
GFA
1,228ft² @ $19 /ft²
Rent $ 23,000
Lohas Park. Phase 7A Montara. Tower 2 (2A) High Floor. Flat E. 2 Rooms
Lohas Park
West
Exclusive
Key
AI Deco & Tour
S.A.
559 ft²
20000
Rent $ 18,500

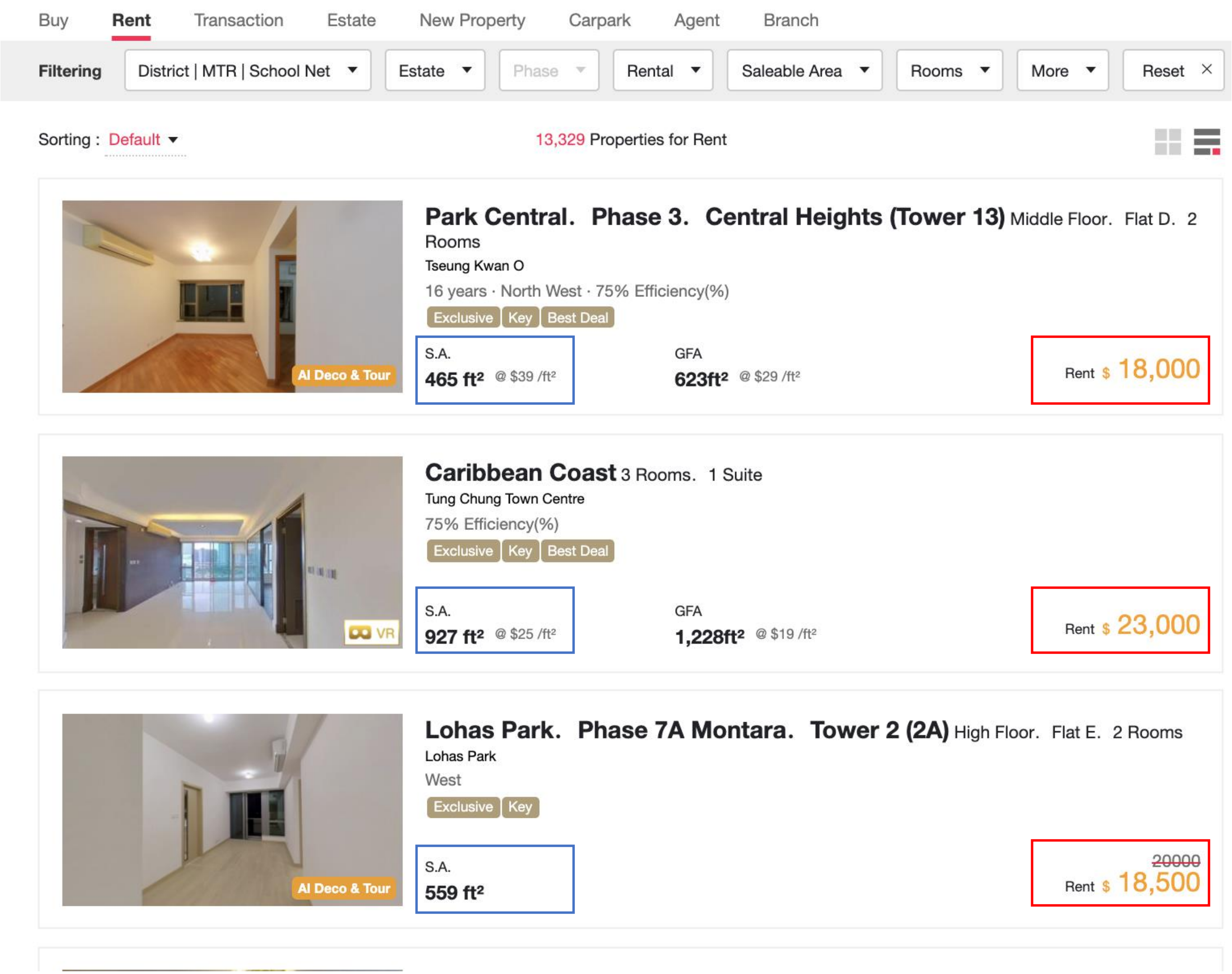
Regression



The price could be affected by

- Living area

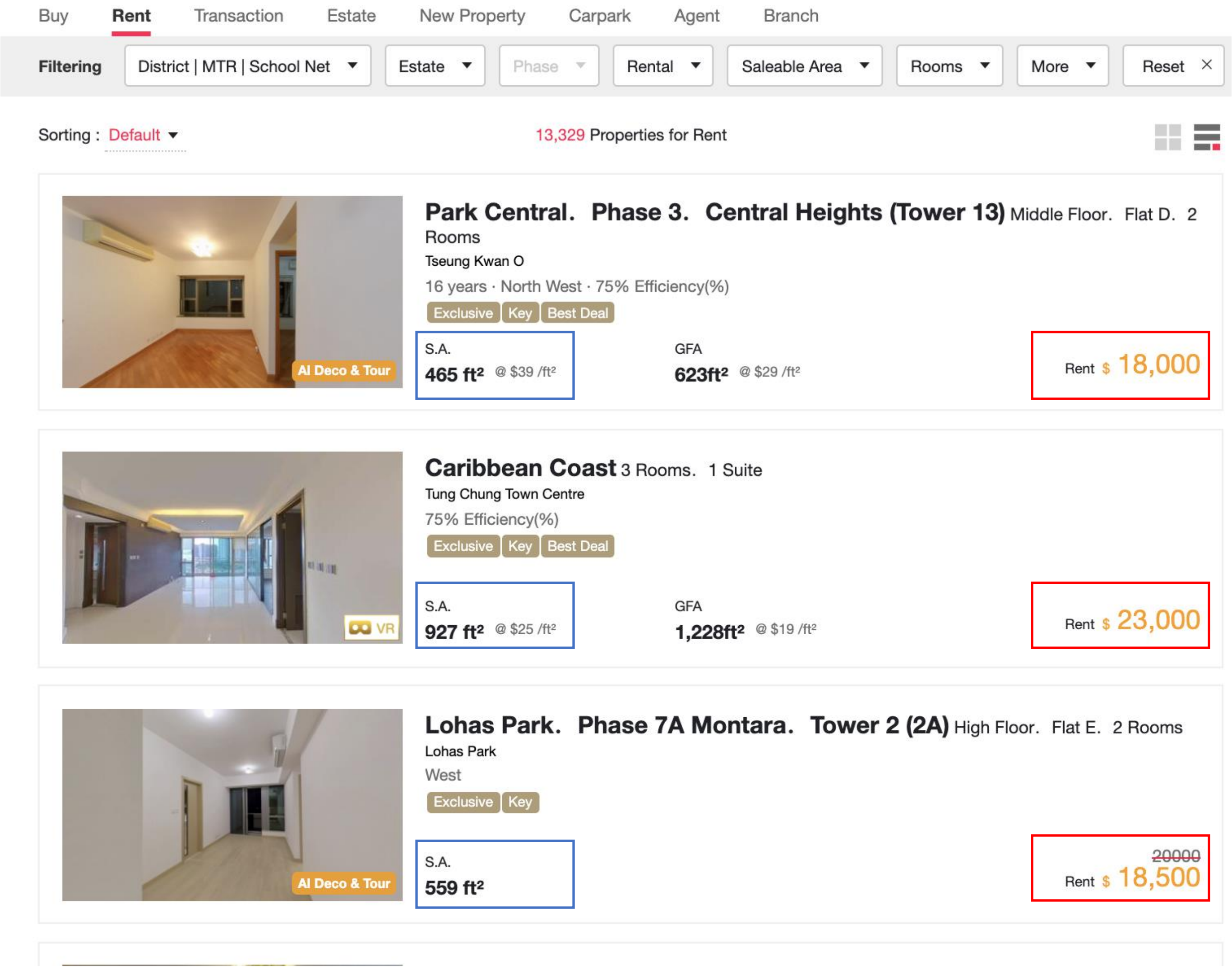
Regression



The price could be affected by

- Living area
- Number of rooms

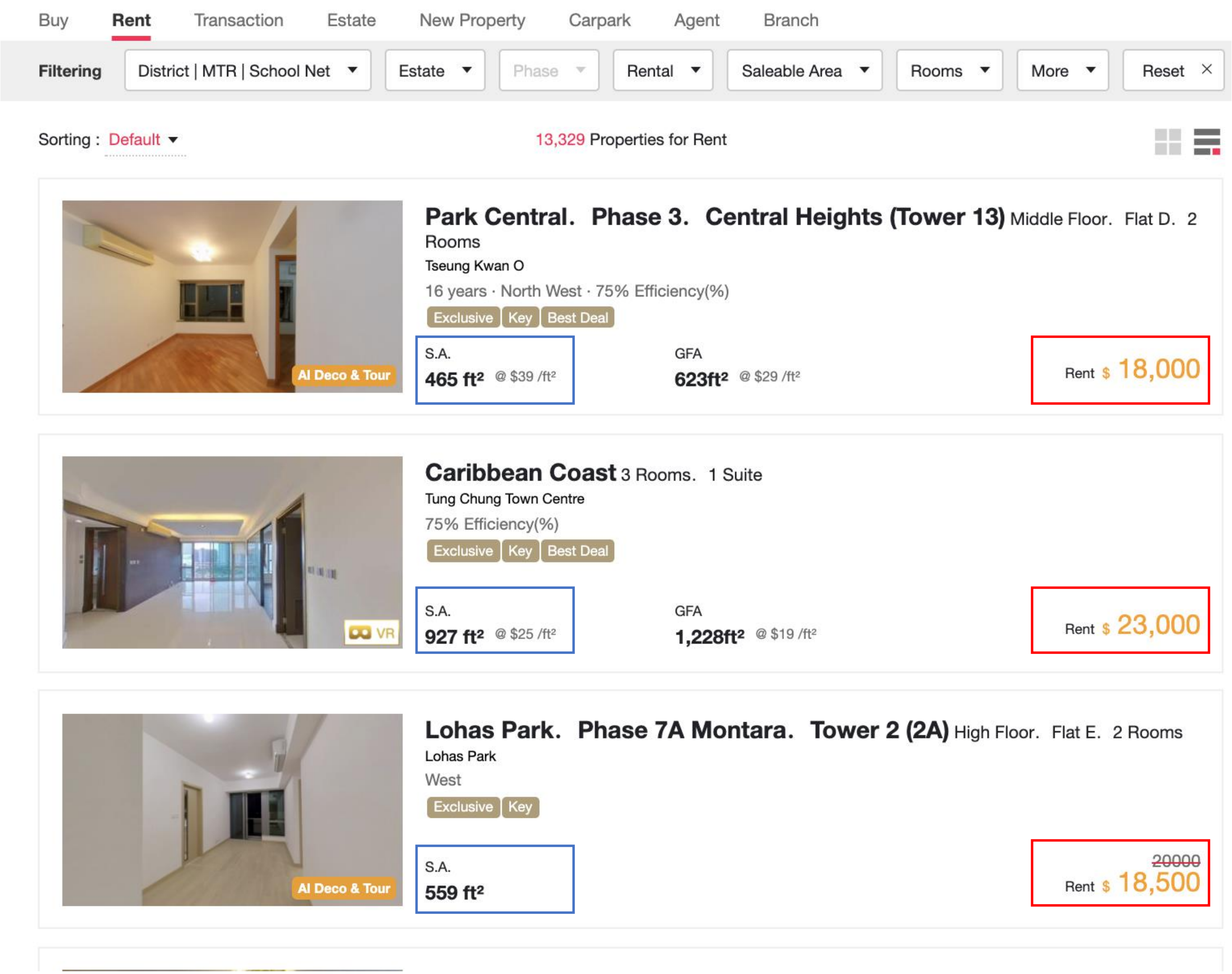
Regression



The price could be affected by

- Living area
- Number of rooms
- Construction time
- …

Regression



The price could be affected by

- Living area
- Number of rooms
- Construction time
- …

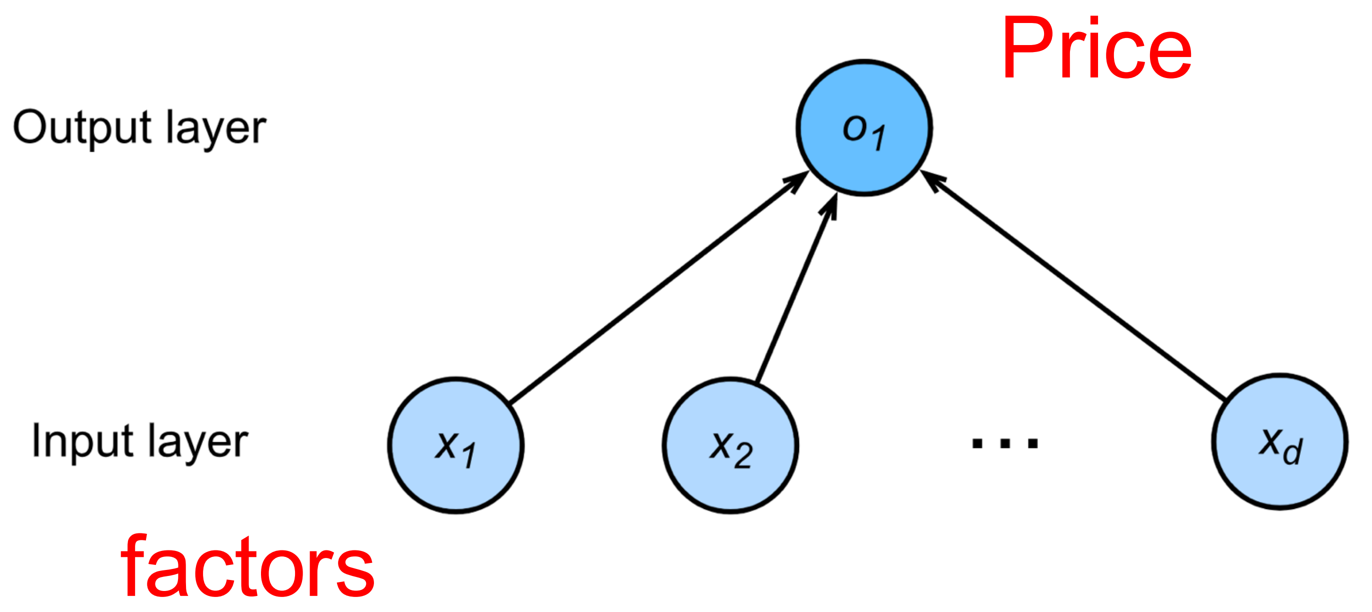
A simplified model:

$$y = w_1 x_1 + w_2 x_2 + w_3 x_3 + b$$

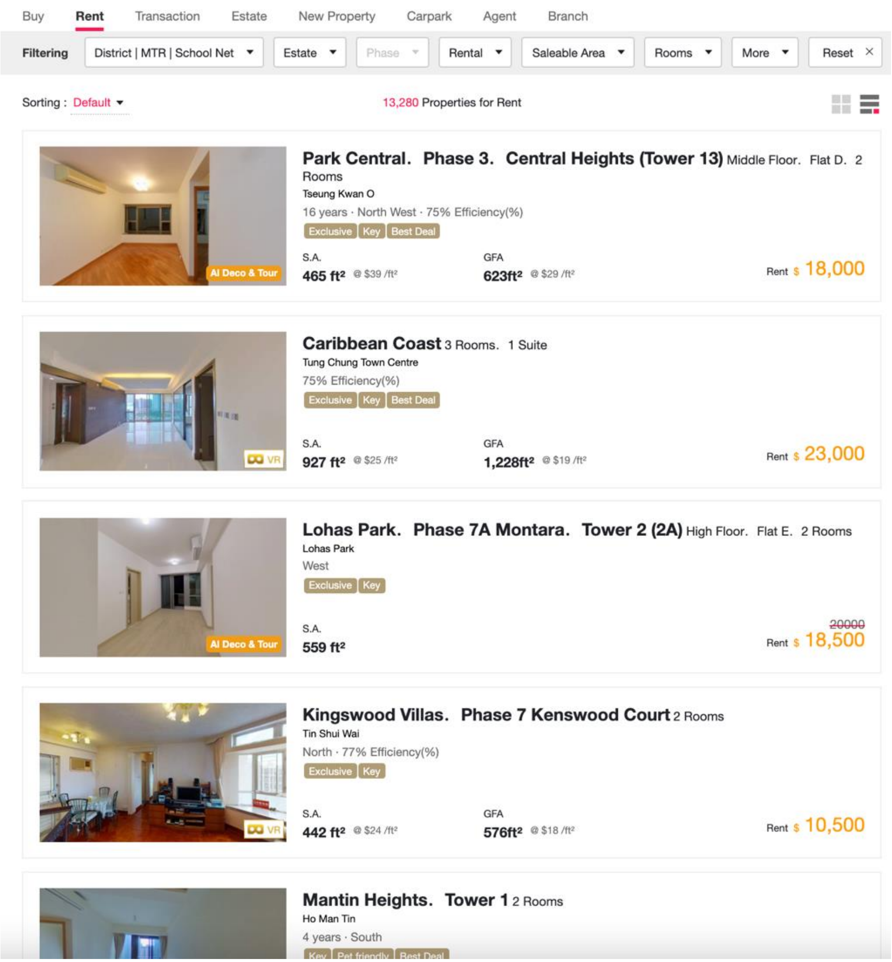
A vector version

$$y = \boldsymbol{w}^T \boldsymbol{x} + b$$

Viewed as a single-layer neural network



Solving $\boldsymbol{w}$ and b



What do we have?

Thousands of data for training.

$$D = \{(x_i, y_i) | i = 1,2, \dots, n\}$$

Defining a loss function.

$$L = \sum_{i=1}^{n} (f(\boldsymbol{x}_i; \boldsymbol{w}, b) - y)^2$$

Loss function: how bad the model fits the data.

Picking $\boldsymbol{w}$ and b with minimal L.

$$\min_{\boldsymbol{w},b} \sum_{i=1}^{n} (f(\boldsymbol{x}_i; \boldsymbol{w}, b) - y)^2$$

Solving $\boldsymbol{w}$ and b

Picking $\boldsymbol{w}$ and b with minimal L.

$$\min_{\boldsymbol{w},b} \sum_{i=1}^{n} (f(\boldsymbol{x}_i; \boldsymbol{w}, b) - y)^2$$

Rewrite in vector form: $\boldsymbol{x} = [x_1, x_2, \dots, x_n] \in R^{d\times n}$

$\boldsymbol{y} = [y_1, y_2, \dots, y_n] \in R^{1\times n}$

$\boldsymbol{w} \in R^{d\times 1}$

$b \in R^1$

$$\min_{\boldsymbol{w},b} \left|\left|\boldsymbol{w}^T\boldsymbol{x} + b - \boldsymbol{y}\right|\right|^2$$

Let $\boldsymbol{w}_a = [\boldsymbol{w}; b] \in R^{(d+1)\times 1}$, $\boldsymbol{x}_a = [\boldsymbol{x}; 1] \in R^{(d+1)\times n}$

A simplified formulation

$$\min_{\boldsymbol{w}_a} \left|\left|\boldsymbol{w}_a^T\boldsymbol{x}_a - \boldsymbol{y}\right|\right|^2$$

Solving $\boldsymbol{w}$ and b

A simplified formulation

$$\min_{\boldsymbol{w}_a} \left|\left|\boldsymbol{w}_a^{\boldsymbol{T}}\boldsymbol{x}_a - \boldsymbol{y}\right|\right|^2$$

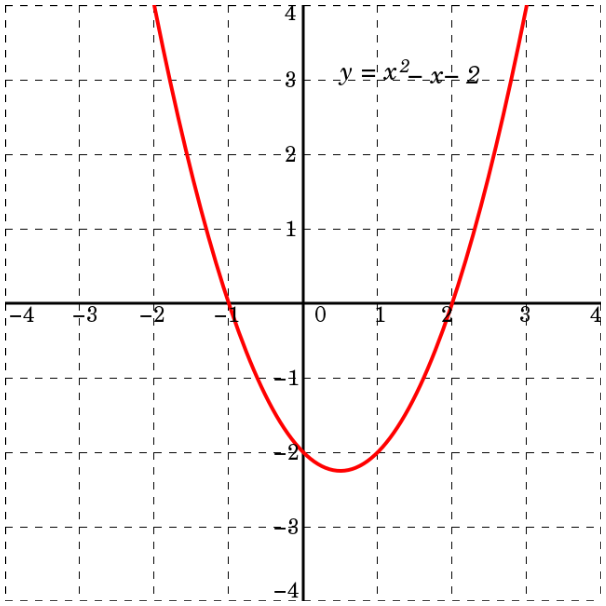
$$\left|\left|\boldsymbol{w}_{\boldsymbol{a}}^{\boldsymbol{T}}\boldsymbol{x}_{\boldsymbol{a}} - \boldsymbol{y}\right|\right|^2$$
$$= \left(\boldsymbol{w}_{\boldsymbol{a}}^{\boldsymbol{T}}\boldsymbol{x}_{\boldsymbol{a}} - \boldsymbol{y}\right)\left(\boldsymbol{w}_{\boldsymbol{a}}^{\boldsymbol{T}}\boldsymbol{x}_{\boldsymbol{a}} - \boldsymbol{y}\right)^T$$
$$= \boldsymbol{w}_a^T\boldsymbol{x}_{\boldsymbol{a}}\boldsymbol{x}_{\boldsymbol{a}}^T\boldsymbol{w}_{\boldsymbol{a}} + \boldsymbol{y}\boldsymbol{y}^T - \boldsymbol{y}\boldsymbol{w}_a^T\boldsymbol{x}_{\boldsymbol{a}} - \boldsymbol{x}_{\boldsymbol{a}}^T\boldsymbol{w}_{\boldsymbol{a}}\boldsymbol{y}$$
$$= \boldsymbol{w}_a^T\boldsymbol{x}_{\boldsymbol{a}}\boldsymbol{x}_{\boldsymbol{a}}^T\boldsymbol{w}_{\boldsymbol{a}} + \boldsymbol{y}\boldsymbol{y}^T - \boldsymbol{2}\boldsymbol{y}\boldsymbol{w}_a^T\boldsymbol{x}_{\boldsymbol{a}}$$

Solving $\boldsymbol{w}$ and b

A simplified formulation

$$\min_{\boldsymbol{w}_a} \left|\left|\boldsymbol{w}_a^T \boldsymbol{x}_a - \boldsymbol{y}\right|\right|^2$$

$$\left|\left|\boldsymbol{w}_a^T \boldsymbol{x}_a - \boldsymbol{y}\right|\right|^2$$
$$= \left(\boldsymbol{w}_a^T \boldsymbol{x}_a - \boldsymbol{y}\right)\left(\boldsymbol{w}_a^T \boldsymbol{x}_a - \boldsymbol{y}\right)^T$$
$$= \boldsymbol{w}_a^T \boldsymbol{x}_a \boldsymbol{x}_a^T \boldsymbol{w}_a + \boldsymbol{y}\boldsymbol{y}^T - \boldsymbol{y}\boldsymbol{w}_a^T \boldsymbol{x}_a - \boldsymbol{x}_a^T \boldsymbol{w}_a \boldsymbol{y}$$
$$= \boldsymbol{w}_a^T \boldsymbol{x}_a \boldsymbol{x}_a^T \boldsymbol{w}_a + \boldsymbol{y}\boldsymbol{y}^T - 2\boldsymbol{y}\boldsymbol{w}_a^T \boldsymbol{x}_a$$



Quadratic function is a convex function.

Optimal solution ⟺ Gradient is 0

Solving $\boldsymbol{w}$ and b

A simplified formulation

$$\min_{\boldsymbol{w}_b} \left|\left|\boldsymbol{w}_a^T \boldsymbol{x}_a - \boldsymbol{y}\right|\right|^2$$

$$\begin{aligned}
&\left|\left|\boldsymbol{w}_a^T \boldsymbol{x}_a - \boldsymbol{y}\right|\right|^2 \\
&= \left(\boldsymbol{w}_a^T \boldsymbol{x}_a - \boldsymbol{y}\right)\left(\boldsymbol{w}_a^T \boldsymbol{x}_a - \boldsymbol{y}\right)^T \\
&= \boldsymbol{w}_a^T \boldsymbol{x}_a \boldsymbol{x}_a^T \boldsymbol{w}_a + \boldsymbol{y}\boldsymbol{y}^T - \boldsymbol{y}\boldsymbol{w}_a^T \boldsymbol{x}_a - \boldsymbol{x}_a^T \boldsymbol{w}_a \boldsymbol{y} \\
&= \boldsymbol{w}_a^T \boldsymbol{x}_a \boldsymbol{x}_a^T \boldsymbol{w}_a + \boldsymbol{y}\boldsymbol{y}^T - 2\boldsymbol{y}\boldsymbol{w}_a^T \boldsymbol{x}_a
\end{aligned}$$

Let $\frac{\partial L}{\partial \boldsymbol{w}_a} = 0$, then we have

$$2\boldsymbol{x}_a \boldsymbol{x}_a^T \boldsymbol{w}_a - 2\boldsymbol{x}_a \boldsymbol{y} = 0$$

A closed form solution

$$\boldsymbol{w}_a = (\boldsymbol{x}_a \boldsymbol{x}_a^T)^{-1} \boldsymbol{x}_a \boldsymbol{y}$$

Let’s see an example!

Regression: PPL

🔍 Overview of the California Housing Dataset

The California Housing Dataset is a classic regression dataset derived from the 1990 California census. It is widely used for educational purposes and for building prototype models to predict **median house prices** in various districts of California.

📊 **Dataset Summary**

• **Number of samples:** 20,640
• **Number of features:** 8 numeric features
• **Target variable:** Median house value (in $100,000)

Feature Name	Description
MedInc	Median income (in $10,000)
HouseAge	Average age of the houses in the district
AveRooms	Average number of rooms per household
AveBedrms	Average number of bedrooms per household
Population	Total population of the block group
AveOccup	Average number of people per household
Latitude	Latitude of the district (geographical coordinate)
Longitude	Longitude of the district (geographical coordinate)

Linear Classification

Classification

The task of the classification algorithm is to find the **mapping function** to map the input (x) to the **discrete** output (y).

Example: Email Spam Detection. The model is trained on millions of emails with different parameters, and whenever it receives a new email, it identifies whether the email is spam or not.

Some Types of Classification Algorithms:

- Logistic Regression (Note: this is a classification method. Why?)
- K-Nearest Neighbors
- Support Vector Machines
- Decision Tree Classification
- Random Forest Classification

Logistic Regression

(1) Linear Component

First, compute a linear combination of input features (like linear regression):

$$z = w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_n x_n$$

- w: Weights (learned from data).
- x: Input features.

(2) Sigmoid Function (Logistic Function)

Map z to **[0, 1]** using the sigmoid:

$$P(y = 1 \mid x) = \sigma(z) = \frac{1}{1 + e^{-z}}$$

- **Interpretation**:
  - $P(y = 1 \mid x)$: Probability of belonging to class 1.
  - $P(y = 0 \mid x) = 1 - P(y = 1 \mid x)$.

(3) Decision Boundary

- If $P(y = 1 \mid x) \geq 0.5$, predict class **1**; otherwise, predict class **0**.

Binary Classification Problem

- IID training samples: $\{(x_1, y_1), \ldots, (x_n, y_n)\}$
- Class label: $y_n \in \{0, 1\}$
- Feature vector: $x_i \in R^d$

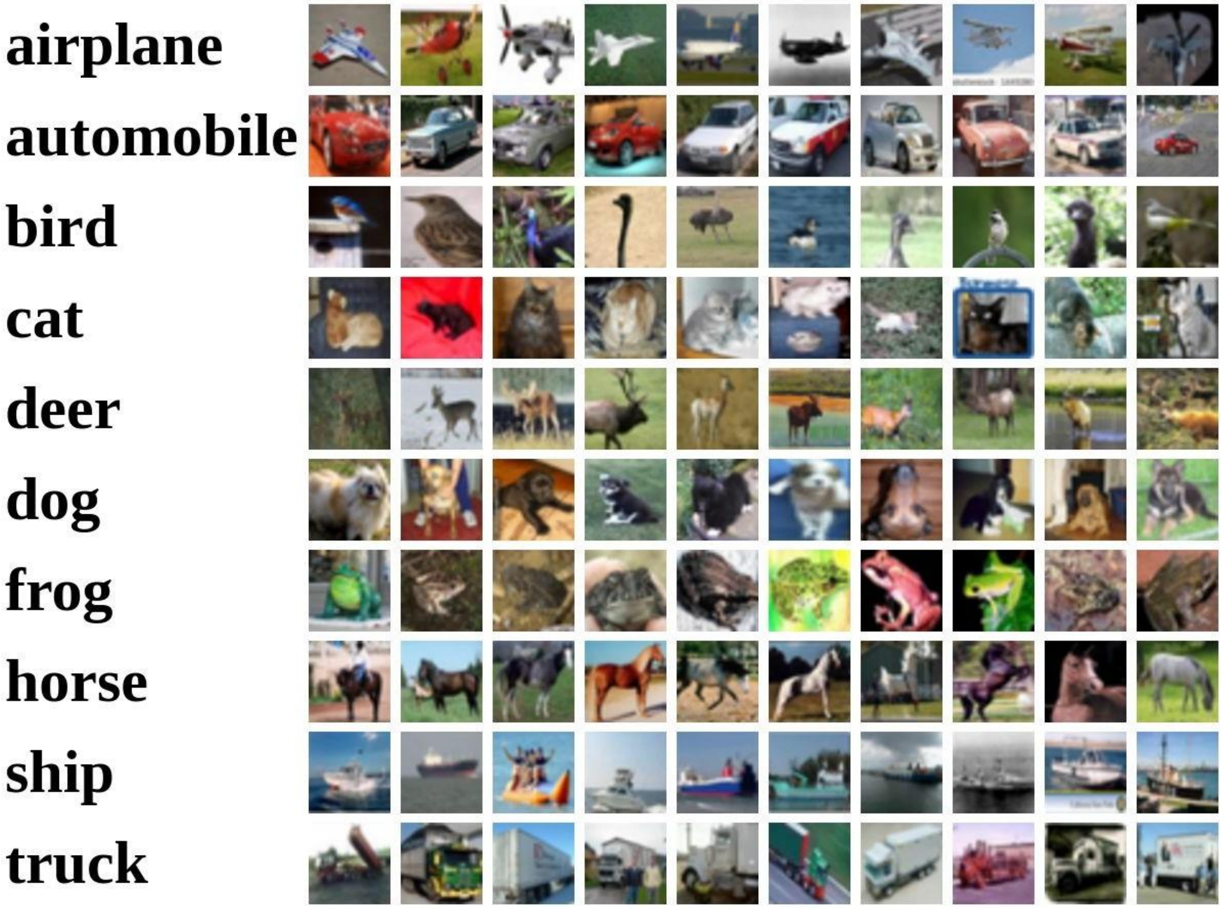
In Binary Classification, the weight vector w is orthogonal to the decision plane. Why?

A linear model $f(\boldsymbol{x}; \boldsymbol{w}) = w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + b = \boldsymbol{wx} + \boldsymbol{b}$.

It is unable to use the value of $f(x; w)$ to predict, then we introduce the decision function $g(.)$, *e.g.*,

$$g\big(f(x; w)\big) = \mathrm{sgn}\big(f(x; w)\big) \triangleq \begin{cases} +1, & \text{if } f(x; w) \geq 0 \\ -1, & \text{if } f(x; w) < 0 \end{cases}.$$

Classification



Example: Cifiar-10 dataset
https://www.cs.toronto.edu/~kriz/cifar.html

CIFAR-10 Dataset

- Contains 60,000 32×32 color images in 10 classes
- 50,000 training images and 10,000 test images
- Classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck

Problem formulation

$\mathrm{I} \in R^{32\times32\times3}$

$\boldsymbol{x} \in R^{3072\times1}$

vectorization

$f(\boldsymbol{x}, \boldsymbol{W}, \boldsymbol{b})$

$\boldsymbol{W}, b$

parameters

10 numbers giving class scores

Prediction: largest score

Problem formulation

$\mathrm{I} \in R^{32\times32\times3}$

$\boldsymbol{x} \in R^{3073\times1}$

vectorization

$f(\boldsymbol{x}, \boldsymbol{W})$

parameters

10 numbers giving class scores

Prediction: largest score

By augmenting the feature, we can remove the bias.

Solving $\boldsymbol{w}$ and b with mult-class SVM loss

cat	**3.2**	1.3	2.2
car	5.1	**4.9**	2.5
frog	-1.7	2.0	**-3.1**

Given a dataset of examples

$$\{x_i, y_i\}_{i=1}^{N}$$

Where x_i is the image and y_i is the label.

$$L = \frac{1}{N}\sum_i L_i(f(x_i, \boldsymbol{W}), y_i)$$

Multiclass SVM loss:

$$L_i = \sum_{j \neq y_i} \begin{cases} 0 & if\ s_{y_i} > s_j + 1 \\ s_j - s_{y_i} + 1 & othersise \end{cases}$$

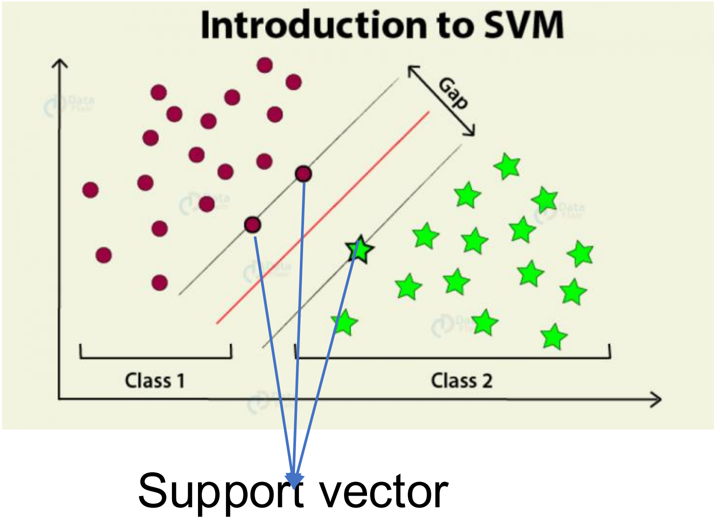
s_{y_i} is the score for correct class.

where $\boldsymbol{s} = f(x_i, \boldsymbol{W}) \in R^3$ is the prediction score.

Support Vector Machine (SVM)

SVM tries to find the **best decision boundary (hyperplane)** that **maximally separates** the data points of different classes.

- In **binary classification**, this means finding a hyperplane that separates the two classes **with the largest margin**.
- The **margin** is the distance between the hyperplane and the closest data points from each class.
- These closest points are called **support vectors**, because they “support” or define the position of the hyperplane.



Solving $\boldsymbol{w}$ and b with mult-class SVM loss



cat	**3.2**	1.3	2.2
car	5.1	**4.9**	2.5
frog	-1.7	2.0	**-3.1**

Multiclass SVM loss:

$$L_i = \sum_{j \neq y_i} \begin{cases} 0 & if\ s_{y_i} > s_j + 1 \\ s_j - s_{y_i} + 1 & othersise \end{cases}$$

If the score value is smaller than that of target class plus a margin

3.2 > -1.7 + 1

Then the loss is 0

Solving $\boldsymbol{w}$ and b with mult-class SVM loss

cat	**3.2**	1.3	2.2
car	5.1	**4.9**	2.5
frog	-1.7	2.0	**-3.1**

Multiclass SVM loss:

$$L_i = \sum_{j \neq y_i} \begin{cases} 0 & if\ s_{y_i} > s_j + 1 \\ s_j - s_{y_i} + 1 & otherwise \end{cases}$$

Otherwise

3.2 < 5.1 + 1

The loss is the difference between the current and target score plus a margin

Solving $\boldsymbol{w}$ and b with mult-class SVM loss

cat	**3.2**	1.3	2.2
car	5.1	**4.9**	2.5
frog	-1.7	2.0	**-3.1**

Multiclass SVM loss:

$$L_i = \sum_{j \neq y_i} \begin{cases} 0 & if\ s_{y_i} > s_j + 1 \\ s_j - s_{y_i} + 1 & othersise \end{cases}$$

Rewrite

$$L_i = \sum_{j \neq y_i} \max\left(0, s_j - s_{y_i} + 1\right)$$

Solving $\boldsymbol{w}$ and b with mult-class SVM loss

Multiclass SVM loss:

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

cat	**3.2**	1.3	2.2
car	5.1	**4.9**	2.5
frog	-1.7	2.0	**-3.1**
Loss	2.9		

= max(0, 5.1 - 3.2 + 1)
+max(0, -1.7 - 3.2 + 1)
= max(0, 2.9) + max(0, -3.9)
= 2.9 + 0
= 2.9

Solving $\boldsymbol{w}$ and b with mult-class SVM loss

cat	**3.2**	1.3	2.2
car	5.1	**4.9**	2.5
frog	-1.7	2.0	**-3.1**
Loss	2.9	0	

Multiclass SVM loss:

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

= max(0, 1.3 - 4.9 + 1)
+max(0, 2.0 - 4.9 + 1)
= max(0, -2.6) + max(0, -1.9)
= 0 + 0
= 0

Solving $\boldsymbol{w}$ and b with mult-class SVM loss

cat	**3.2**	1.3	2.2
car	5.1	**4.9**	2.5
frog	-1.7	2.0	**-3.1**
Loss	2.9	0	12.9

Multiclass SVM loss:

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

= max(0, 2.2 - (-3.1) + 1)
+max(0, 2.5 - (-3.1) + 1)
= max(0, 6.3) + max(0, 6.6)
= 6.3 + 6.6
= 12.9

Solving $\boldsymbol{w}$ and b with mult-class SVM loss



cat	**3.2**	1.3	2.2
car	5.1	**4.9**	2.5
frog	-1.7	2.0	**-3.1**
Loss	2.9	0	12.9

Multiclass SVM loss:

$$L_i = \sum_{j \neq y_i} \max\left(0, s_j - s_{y_i} + 1\right)$$

$$L = \frac{1}{N} \sum_{i=1}^{N} L_i$$

$$L = (2.9 + 0 + 12.9)/3$$
$$= \mathbf{5.27}$$

How about using CE loss?

Cross–Entropy Loss

For multi–class classification with K classes (K=10 for CIFAR–10):

```
CE_loss = −Σ(y_i * log(p_i))
```

Where:

- y_i is the true label (one–hot encoded)
- p_i is the predicted probability for class i
- The sum is over all classes

Let’s see an example!

Classification on CIFAR-10

CIFAR-10 Dataset

- **Task:** Image classification
- **Number of classes:** 10
- **Number of images:** 60,000
    - 50,000 training images
    - 10,000 test images
- **Image size:** 32×32 pixels
- **Channels:** 3 (RGB)
- **Classes:**
    - airplane
    - automobile
    - bird
    - cat
    - deer
    - dog
    - frog
    - horse
    - ship
    - truck

Each class has exactly 6,000 images.

📋 **CIFAR-100 Dataset**

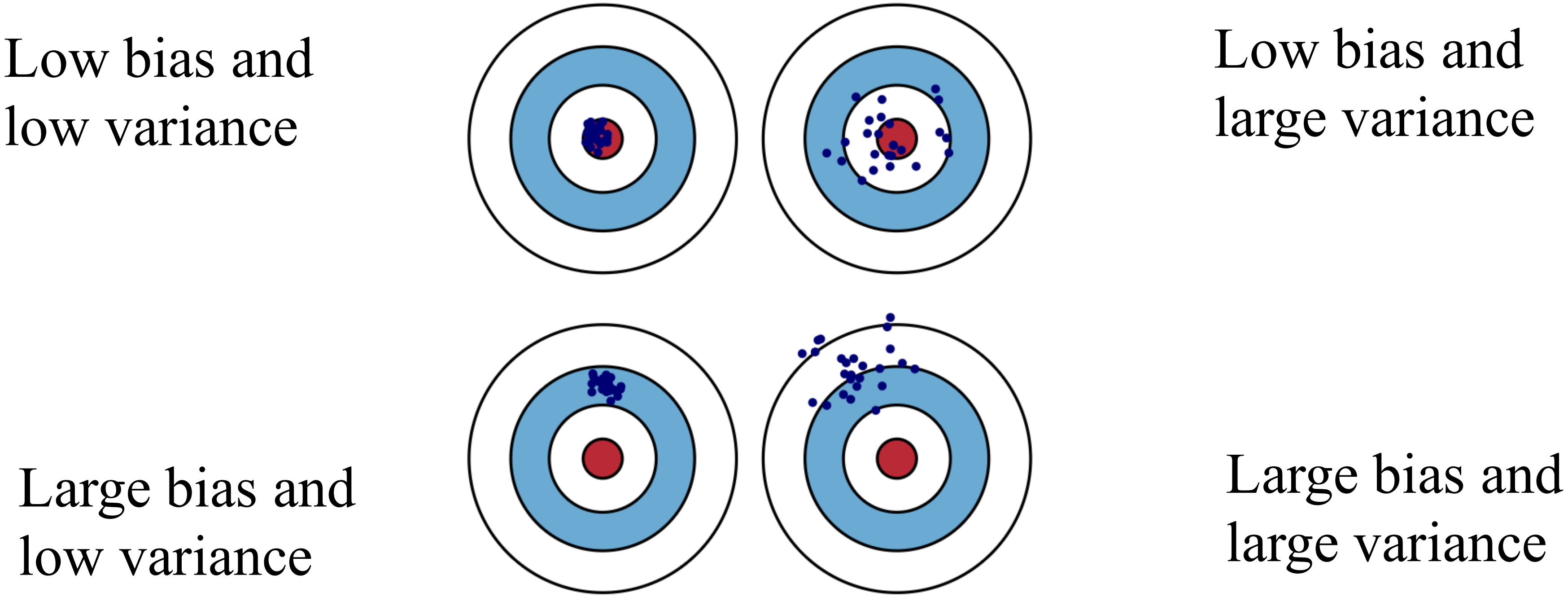
- **Task:** Fine-grained image classification
- **Number of classes:** 100
- **Number of images:** 60,000
    - 50,000 training images
    - 10,000 test images
- **Image size:** 32×32 pixels
- **Channels:** 3 (RGB)
- **Structure:**
    - 100 **fine** classes (each with 600 images)
    - Grouped into 20 **coarse** superclasses (e.g., aquatic mammals, vehicles 1, large carnivores, etc.)

Comparison between Regression and Classification

Regression Algorithm	Classification Algorithm
The task is to map the input value (x) with the **continuous** output variable (y).	The task is to map the input value (x) with the **discrete** output variable (y).
We try to find the **best fit line**, which can predict the output more accurately.	We try to find the **decision boundary**, which can divide the dataset into different classes.
can be further divided into Linear and Non-linear Regression.	can be divided into Binary Classifier and Multi-class Classifier.
$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(f_i - o_i)^2}$	$\boxed{\text{Accuracy, Sensitivity, Specificity, ROC, } etc.}$

Bias and Variance

- **Bias:** the difference between the average **prediction of model** and the **correct value** which we are trying to predict.
- **Variance:** Model with high variance pays a lot of attention to **training data** and does not generalize on the **test data**.



Bias-Variance Tradeoff

- Assume that we have minimized the error (*i.e.*, loss) with respect to **training data**
  - Low **training** error does not imply good expected performance on testing data: **overfitting**
- We would like to reason about the **expected loss** over:

  Training Data: $\{(x_1, y_1), \ldots, (x_n, y_n)\}$

  - Test point: (x_*, y_*)
- We will decompose the expected loss into:

$$\boxed{\mathbf{E}_{D,(y_*,x_*)}\left[\left(y_* - f(x_*|D)\right)^2\right] = \text{Noise} + \text{Bias}^2 + \text{Variance}}$$

- Define (**unobserved**) the **true** model (h):

$$y_* = h(x_*) + \epsilon_*$$

Assume 0 mean noise
[bias goes in $h(x_*)$]

- Completed the squares with: $\boxed{h(x_*) = h_*}$

$$\mathbf{E}_{D,(y_*,x_*)}\left[\left(y_* - f(x_*|D)\right)^2\right]$$ Expected Loss

$$= \mathbf{E}_{D,(y_*,x_*)}\left[\left(\underbrace{y_* - h(x_*)}_{a} + \underbrace{h(x_*) - f(x_*|D)}_{b}\right)^2\right]$$

$$(a+b)^2 = a^2 + b^2 + 2ab$$

$$= \mathbf{E}_{\epsilon_*}\left[\left(y_* - h(x_*)\right)^2\right] + \mathbf{E}_D\left[\left(h(x_*) - f(x_*|D)\right)^2\right]$$
$$+ 2\mathbf{E}_{D,(y_*,x_*)}\left[y_*h_* - y_*f_* - h_*h_* + h_*f_*\right]$$

- Define (unobserved) the true model (h):

$$y_* = h(x_*) + \epsilon_*$$

- Completed the squares with: $h(x_*) = h_*$

$$\mathbf{E}_{D,(y_*,x_*)}\left[(y_* - f(x_*|D))^2\right]$$ Expected Loss

$$= \mathbf{E}_{D,(y_*,x_*)}\left[(y_* - h(x_*) + h(x_*) - f(x_*|D))^2\right]$$

$$= \mathbf{E}_{\epsilon_*}\left[(y_* - h(x_*))^2\right] + \mathbf{E}_D\left[(h(x_*) - f(x_*|D))^2\right]$$

$$\underbrace{\cancel{+ 2\mathbf{E}_{D,(y_*,x_*)}\left[y_* h_* - y_* f_* - h_* h_* + h_* f_*\right]}}_{\text{Substitute defn. } y_* = h_* + \epsilon_*}$$

$$\mathbf{E}\left[(h_* + \epsilon_*)h_* - (h_* + \epsilon_*)f_* - h_* h_* + h_* f_*\right] =$$

$$\cancel{h_* h_*} + \cancel{\mathbf{E}[\epsilon_*]\, h_*} - \cancel{h_* \mathbf{E}[f_*]} - \cancel{\mathbf{E}[\epsilon_*]\, f_*} - \cancel{h_* h_*} + \cancel{h_* \mathbf{E}[f_*]}$$

- Define (unobserved) the true model (h):

$$y_* = h(x_*) + \epsilon_*$$

- Completed the squares with: $h(x_*) = h_*$

$$\mathbf{E}_{D,(y_*,x_*)}\left[(y_* - f(x_*|D))^2\right] \quad \text{Expected Loss}$$

$$= \mathbf{E}_{D,(y_*,x_*)}\left[(y_* - h(x_*) + h(x_*) - f(x_*|D))^2\right]$$

$$= \underbrace{\mathbf{E}_{\epsilon_*}\left[(y_* - h(x_*))^2\right]}_{\substack{\text{Noise Term} \\ \text{(out of our control)}}} + \boxed{\underbrace{\mathbf{E}_D\left[(h(x_*) - f(x_*|D))^2\right]}_{\substack{\text{Model Estimation Error} \\ \text{(we want to minimize this)}}}}$$

☹

Expand

- Minimum error is governed by the noise.

- Expanding on the model estimation error:

$$\mathbf{E}_D\left[(h(x_*)-f(x_*|D))^2\right]$$

- Completing the squares with $\mathbf{E}\left[f(x_*|D)\right]=\bar{f}_*$

$$\begin{aligned}
&\mathbf{E}_D\left[(h(x_*)-f(x_*|D))^2\right]\\
&\quad=\mathbf{E}\left[(h(x_*)-\mathbf{E}\left[f(x_*|D)\right]+\mathbf{E}\left[f(x_*|D)\right]-f(x_*|D))^2\right]\\
&\quad=\mathbf{E}\left[(h(x_*)-\mathbf{E}\left[f(x_*|D)\right])^2\right]+\mathbf{E}\left[(f(x_*|D)-\mathbf{E}\left[f(x_*|D)\right])^2\right]\\
&\qquad\quad+\cancel{2\mathbf{E}\left[h_*\bar{f}_*-h_*f_*-\bar{f}_*f_*+\bar{f}_*^2\right]}
\end{aligned}$$

$$=h_*\bar{f}_*-h_*\mathbf{E}\left[f_*\right]-\bar{f}_*\mathbf{E}\left[f_*\right]+\bar{f}_*^2=$$

$$h_*\bar{f}_*-h_*\bar{f}_*-\bar{f}_*\bar{f}_*+\bar{f}_*^2=0$$

- Expanding on the model estimation error:

$$\mathbf{E}_D\left[(h(x_*) - f(x_*|D))^2\right]$$

- Completing the squares with $\mathbf{E}\left[f(x_*|D)\right] = \bar{f}_*$

$$\begin{aligned}
&\mathbf{E}_D\left[(h(x_*) - f(x_*|D))^2\right] \\
&\quad = \underbrace{(h(x_*) - \mathbf{E}\left[f(x_*|D)\right])^2}_{(\text{Bias})^2} + \underbrace{\mathbf{E}\left[(f(x_*|D) - \mathbf{E}\left[f(x_*|D)\right])^2\right]}_{\text{Variance}}
\end{aligned}$$

- Tradeoff between bias and variance:
    - **Simple Models:** High Bias, Low Variance $\Rightarrow$ Underfitting
    - **Complex Models:** Low Bias, High Variance $\Rightarrow$ Overfitting

Bias-Variance Plot

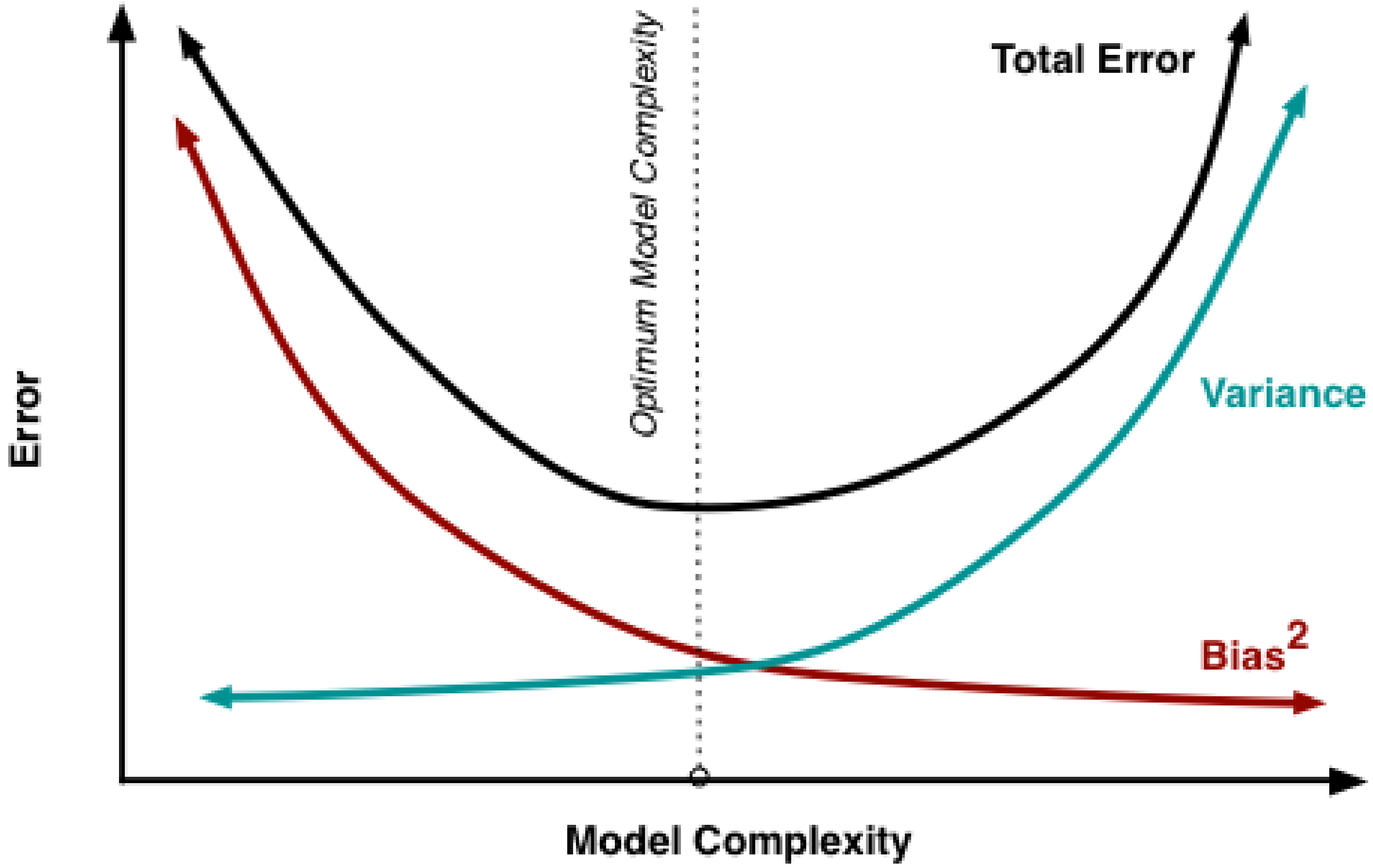
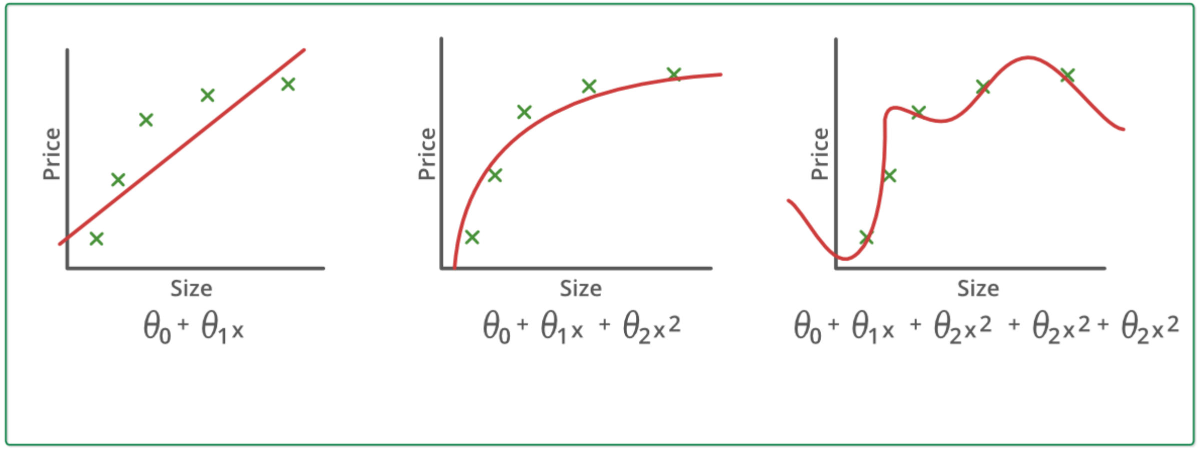
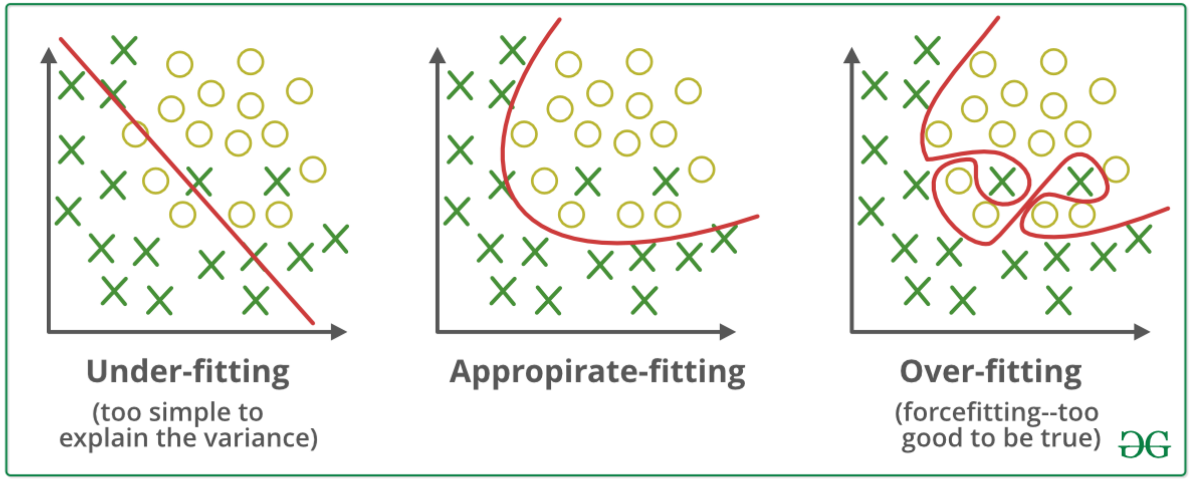


Image from http://scott.fortmann-roe.com/docs/BiasVariance.html

Overfitting and underfitting

For that, we have overfitting and underfitting, which are majorly responsible for the poor performances of the machine learning algorithms.





Underfitting

- It cannot capture the underlying trend of the data.
- It destroys the accuracy of our machine learning model.
- **It usually happens when we have less data to build an accurate model and also when we try to build a linear model for a non-linear data.**

Underfitting – High bias and low variance

Techniques to reduce underfitting :

1. Use more data;
2. Increase model complexity;
3. Increase number of features, performing feature engineering;
4. Remove noise from the data;
5. Increase #epochs or increase the duration of training.

Overfitting

- Limited training data
- Trains the model with **a lot of data (**when a model gets trained with so much of data, it starts learning from the noise and inaccurate data in the dataset).
- High model complexity.

Overfitting – High variance and low bias

Techniques to reduce overfitting:
1. Increase training data; training data purification;
2. Reduce model complexity;
3. Early stopping during the training phase (have an eye over the loss over the training; as soon as loss begins to increase, stop training);
4. Use dropout for DNNs.

Evaluation Metrics

Evaluation metrics are **quantitative measures** used to evaluate the performance or effectiveness of a system, model, algorithm, or process.

Why should we care?

- Imagine you have 2 different probabilistic classification models
  - *e.g.*, logistic regression *vs.* neural network

- How do you know which one is better?

- Can you provide quantitative evidence beyond a gut feeling and subjective interpretation?

Recall Basics

		Model Prediction	
		It's NOT a Heart Attack	*Heart Attack!!!*
Gold Standard Truth	*Was NOT a Heart Attack*	*A*	*B*
	Was a Heart Attack	*C*	*D*

Some Terms: confusion matrix

		Model Prediction	
		No Event (阴性)	*Event (阳性)*
Gold Standard Truth	*No Event (阴性)*	*True Negative*	*False Positive (Type 1 Error)*
	Event (阳性)	*False Negative (Type 2 error)*	*True Positive*

假阳性

假阴性

Which one do you think is more severe？

Accuracy

$$Acc = \frac{number\ of\ hits}{total\ number} \times 100\%$$

Although it can judge the overall correct rate, it is not a **good** indicator to measure the results in the case of **imbalanced samples** (however, this situation often occurs).

For example, there is a model for detecting COVID-19 pneumonia. Using this model to investigate COVID-19 pneumonia in a certain community, there are 280,000 normal cases (no infection) and 50 cases of COVID-19 pneumonia. The model judges everyone as normal **No infection**, that is, **no patients** with COVID-19 pneumonia were found, and the accuracy rate at this time was 280,000/280050 = 99.9%. Although the accuracy rate at this time is high, the model did not find COVID-19 patients, so the model is actually very **bad**.

ROC Curves

- Receiver operator characteristic
- Models ability to distinguish between false & true positives
- AUC (Area Under the Curve)

Use Multiple Contingency Tables

- *TRUE POSITIVE RATE* (also called *Sensitivity*)

$$\textit{Sensitivity} = \frac{\textbf{(True Positives)}}{\textbf{(True Positives) + (False Negatives)}}$$

- *FALSE POSITIVE RATE* (also called *Specificity*)

$$\textit{Specificity} = \frac{\textbf{(True Negatives)}}{\textbf{(False Positives) + (True Negatives)}}$$

- Plot Sensitivity *vs.* (1 – Specificity) False Positive Rate

		Model Prediction	
		Neg	*Pos*
Gold Standard Truth	*Neg*	*TN*	*FP*
	Pos	*FN*	*TP*

Sensitivity

- Proportion of **patients with disease** who are tested positive with a test；
- A 100% sensitive test will not have any false negative results；
- Therefore, a negative result of a highly sensitive test means it is likely to be **a true negative (it rules out the disease)**.

$$Sensitivity = \frac{\text{(True Positives)}}{\text{(True Positives) + (False Negatives)}}$$

Specificity

- Proportion of **patients without disease** who are tested negative with a test
- A 100% specific test will not have false positive results (although it may have high rate of false negative results)
- Therefore, a positive result of a highly specific test means it is likely **to be true positive**

$$Specificity = \frac{\text{(True Negatives)}}{\text{(False Positives)} + \text{(True Negatives)}}$$

Example

Problem and Data (Breast cancer)

Diagnosis Result	Surgical pathology results		Total
	Patient (P)	Non-patient (N)	
Positive	80	100	180
Negative	20	800	820
Total	100	900	1000

Compute the sensitivity and specificity.

Example

Problem and Data (Breast cancer)

Diagnosis Result	Surgical pathology results		Total
	Patient (P)	Non-patient (N)	
Positive (P)	80 (TP)	100 (FP)	180
Negative (N)	20 (FN)	800 (TN)	820
Total	100	900	1000

Sen: 80%(TP/(TP+FN)) Sp: 89% (TN/(TN+FP))

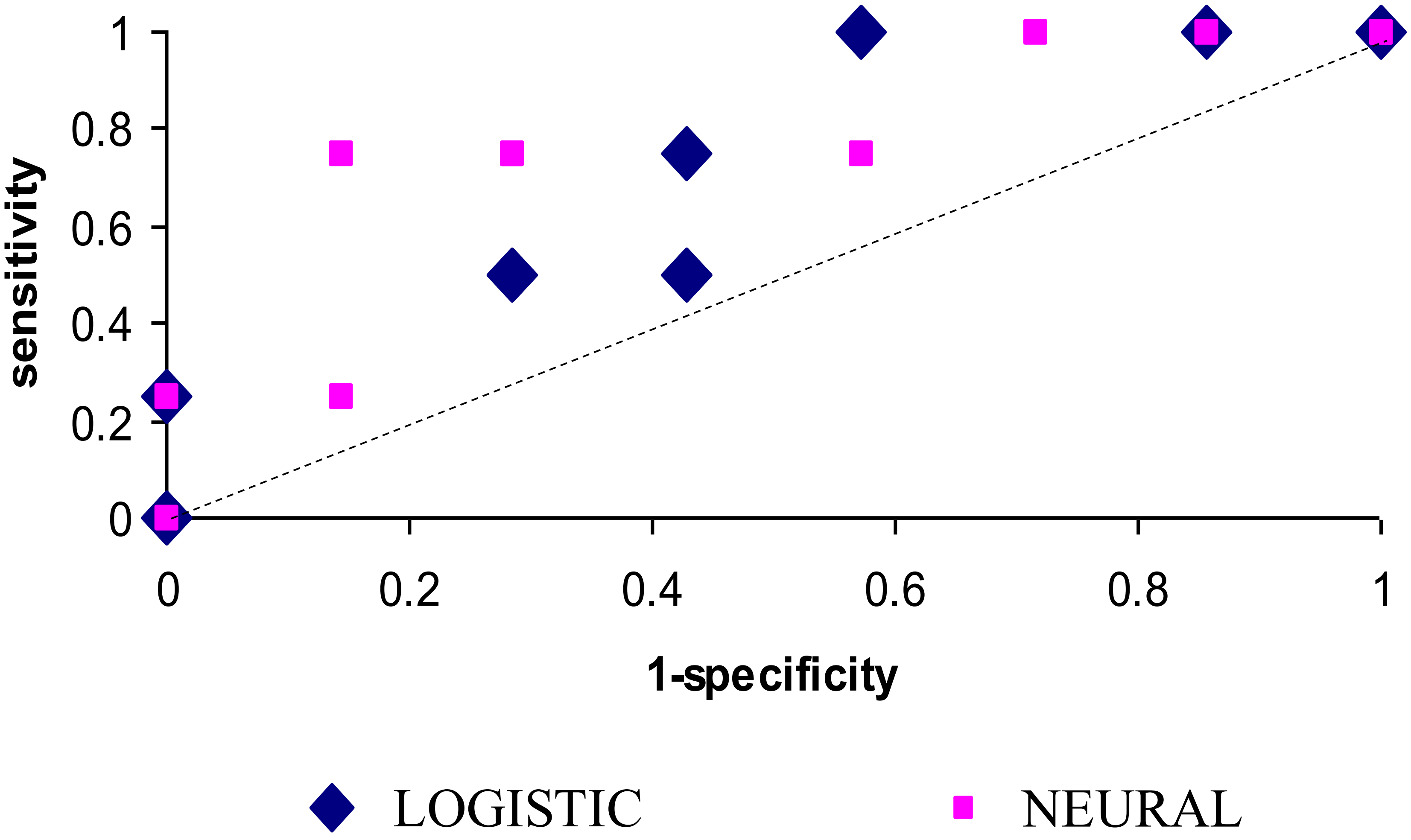
		Model Prediction	
		Neg	*Pos*
Gold Standard Truth	*Neg*	*TN*	*FP*
	Pos	*FN*	*TP*

Summary

- If the sensitivity of a diagnostic test is relatively low, there will be many false-negative patients. This will delay the patient's visit, affect the development and recovery of the disease, and even lead to premature death of the patient.

- If the specificity of a diagnostic test is relatively low, there will be many false positive patients. This will waste medical resources and cause unprovoked panic and anxiety among patients.

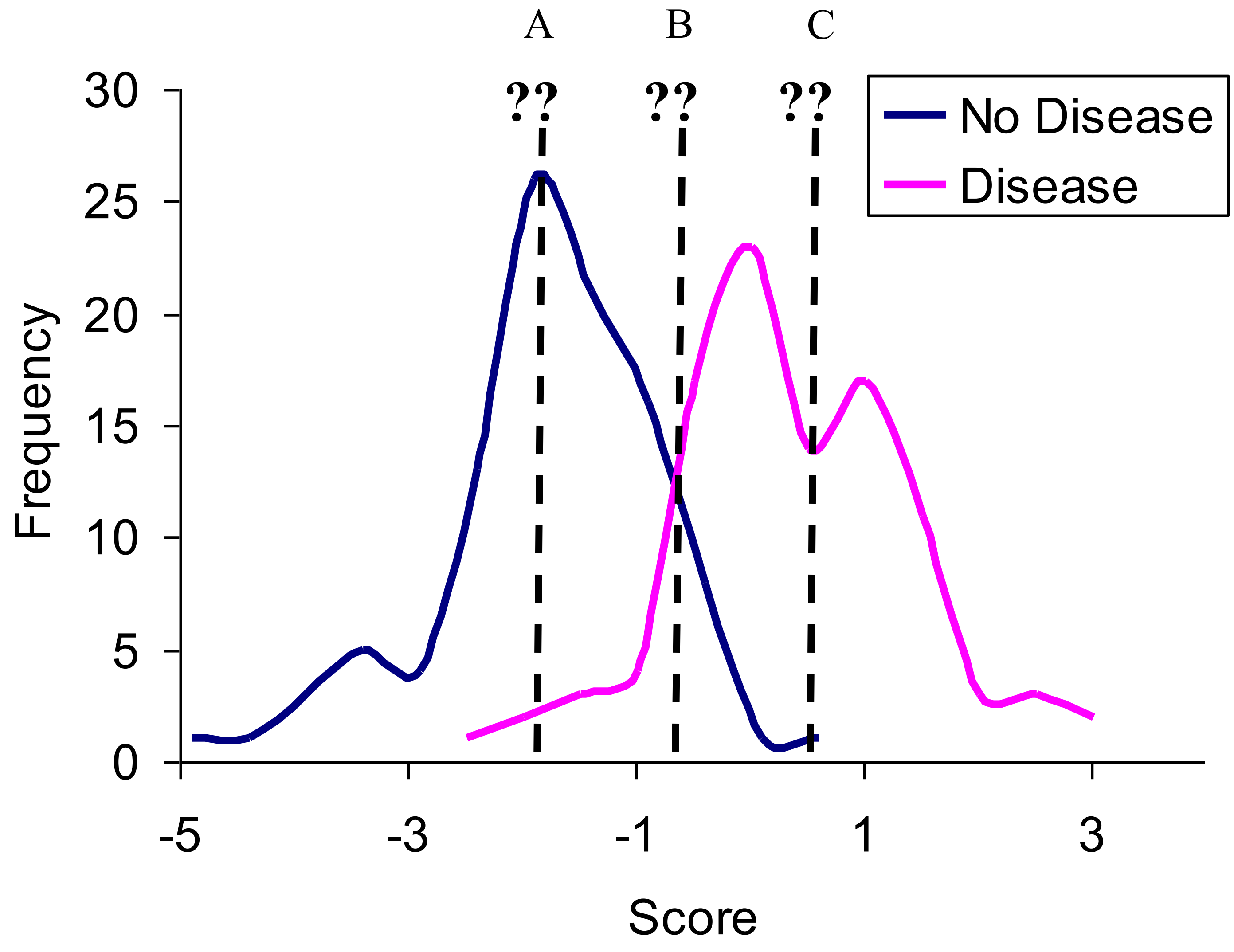
ROC Plot



The idea case:

- It would be on the top-left corner of the ROC graph corresponding to the coordinate (0, 1) in the cartesian plane.
- Both the Sensitivity and Specificity, would be the highest and the classifier would correctly classify all the Positive and Negative class points.

Visual Perspective



How to plot ROC? Let' s see an example

Suppose now we have 20 samples, **Class is the ground truth**, and **Score is the probability of the positive sample given by the classifier**.

Inst#	Class	Score	Inst#	Class	Score
1	p	.9	11	p	.4
2	p	.8	12	n	.39
3	n	.7	13	p	.38
4	p	.6	14	n	.37
5	p	.55	15	n	.36
6	p	.54	16	n	.35
7	n	.53	17	p	.34
8	n	.52	18	n	.33
9	p	.51	19	p	.30
10	n	.505	20	n	.1

1. Sort by score from largest to smallest
2. Set all the scores as a threshold, and then the labels of these 20 samples will change. When its score is greater than or equal to the current threshold, it is a positive sample, otherwise it is no sample.
3. In this way, for each threshold, one set of FPR and TPR can be calculated. The example can be a total of 20 sets.
4. When the threshold is set to 1 and 0, two points (0,0) and (1,1) on the ROC curve can be obtained.

Let’s see an example!

ROC plot

How to plot ROC if it is not a binary classifier problem?

The ROC curve is for binary classifications. There are also methods for obtaining the ROC curve for **multi-class classification** problems:

Suppose the number of test samples is m and the number of classes is n.

- After the training is completed, calculate the probability or confidence of each test sample in each category, and obtain a matrix P of size $[m, n]$, where each row represents the probability value of a test sample in each category (according to the category label sort).
- Correspondingly, the label of each test sample is converted into a binary-like form, and each position is used to mark whether it belongs to the corresponding class, which can also obtain an $[m, n]$ size label matrix L.

Method:

- Under each class, the probability of m test samples being that category (column in matrix P) can be obtained.
- Therefore, according to each corresponding column in the probability matrix P and the label matrix L, the false positive rate (FPR) and the true rate (TPR) under **each threshold** can be calculated to draw a ROC curve.
- In this way, a total of n ROC curves can be drawn. Finally, **average** the n ROC curves to get the final ROC curve.

Precision and Recall

Precision score is the number of true positive results divided by the number of all positive results.

Recall score, also known as *Sensitivity* or *true positive rate*, is the number of true positive results divided by the number of all samples that should have been identified as positive.

$$\text{Precision} = \frac{TP}{TP+FP}$$

$$\text{Recall} = \frac{TP}{TP+FN}$$

		Model Prediction	
		No Event (阴性)	*Event (阳性)*
Gold Standard Truth	*No Event (阴性)*	*TN*	*FP*
	Event (阳性)	*FN*	*TP*

$$\text{Specificity} = \frac{TN}{TN+FP}$$

Thank you!

Questions?

Why IID?

Deep Neural Networks (DNNs) assume that training data is **Independent and Identically Distributed (IID)**. This assumption is critical for ensuring **generalization** and **stable optimization**.

1. Definition of IID

- **Independent**: Samples are statistically unrelated (e.g., coin flips do not influence each other).
- **Identically Distributed**: All samples are drawn from the same underlying probability distribution.

Mathematically:

For data $\{x_1, x_2, \ldots, x_n\}$, IID implies:

$$P(x_1, x_2, \ldots, x_n) = \prod_{i=1}^{n} P(x_i), \quad P(x_i) = P(x_j) \quad \forall i, j.$$

Why IID?

(1) Generalization Performance

- Non-IID data breaks these theoretical guarantees, leading to overfitting or underfitting.

(2) Batch Normalization (BatchNorm): Batch statistics (mean/variance) estimate the global data distribution.

- If data is non-IID (e.g., correlated samples in a batch), BatchNorm introduces bias.

(3) Optimization Stability

• **Non-independent Data**: Gradient updates are dominated by local samples (e.g., similar consecutive images), causing oscillations or local optima.

• **Non-identical Data**: Subpopulations may be overfitted while others are ignored.